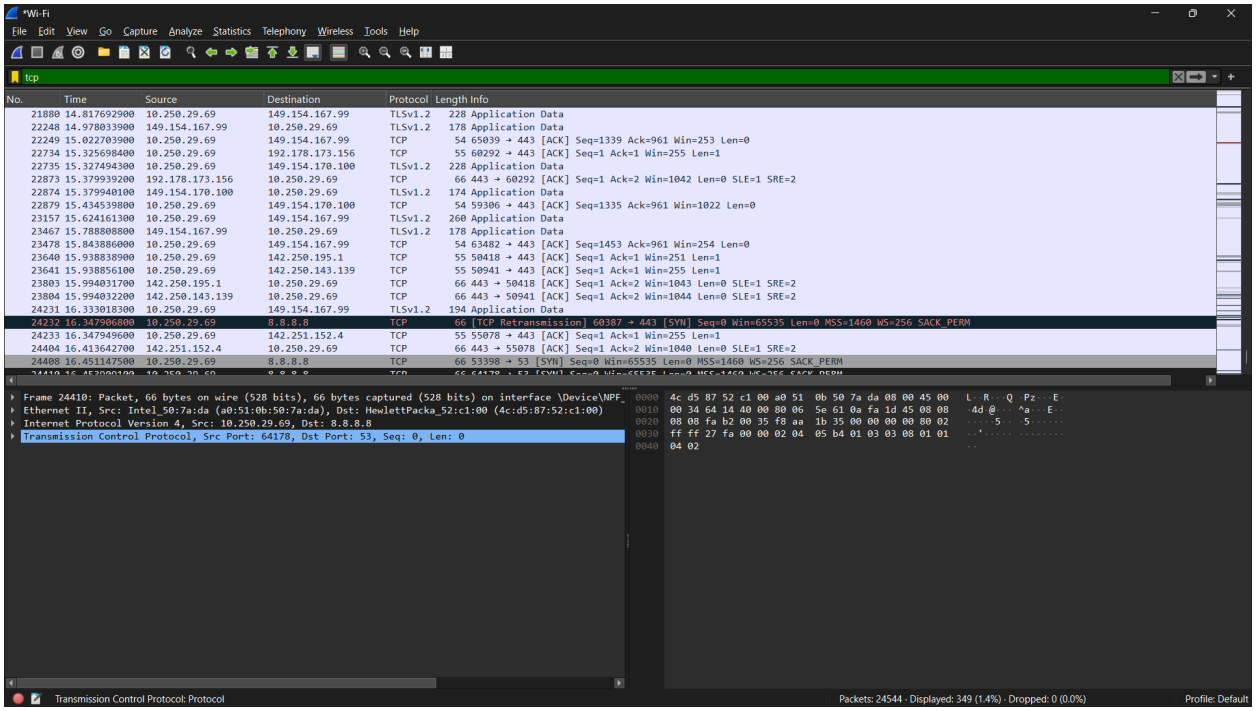
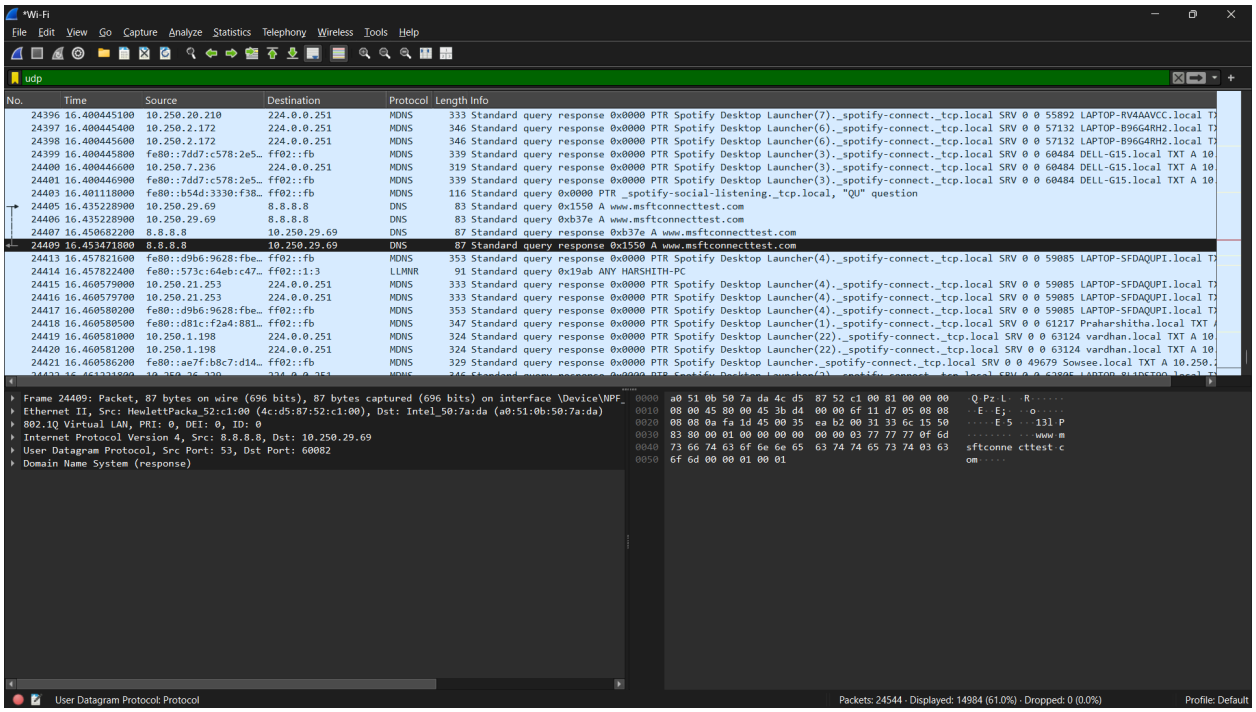


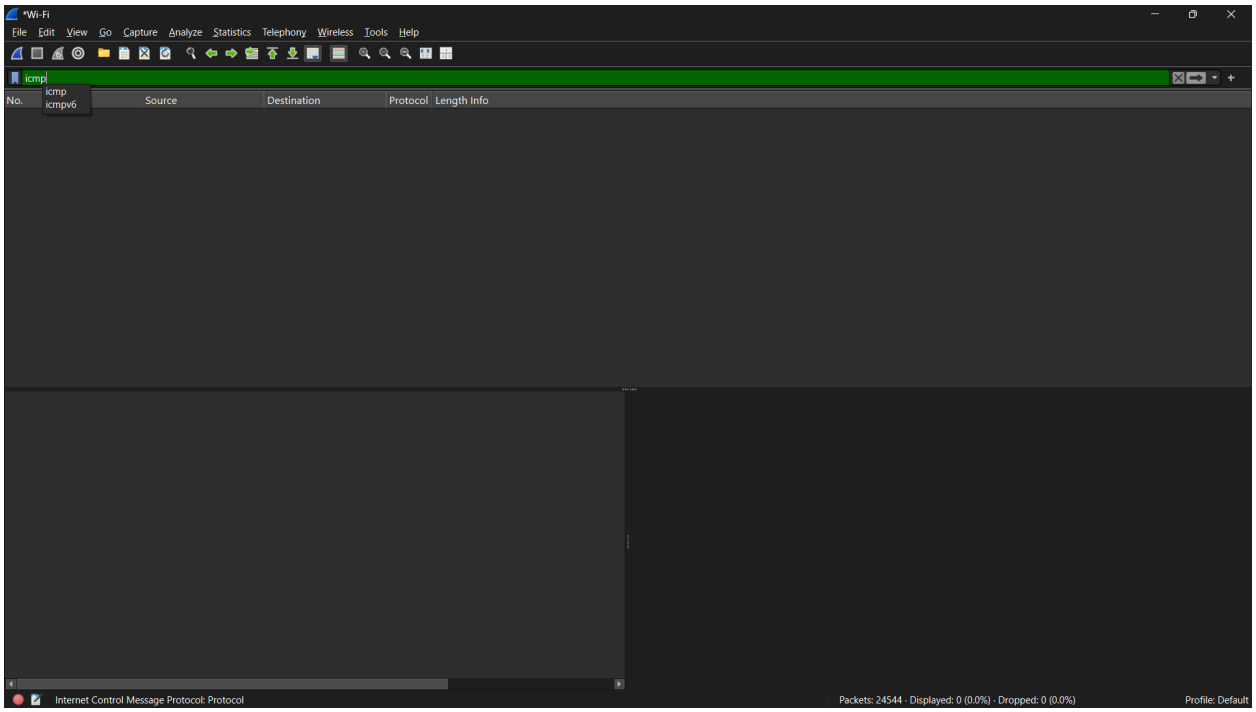
Total packets: 24544



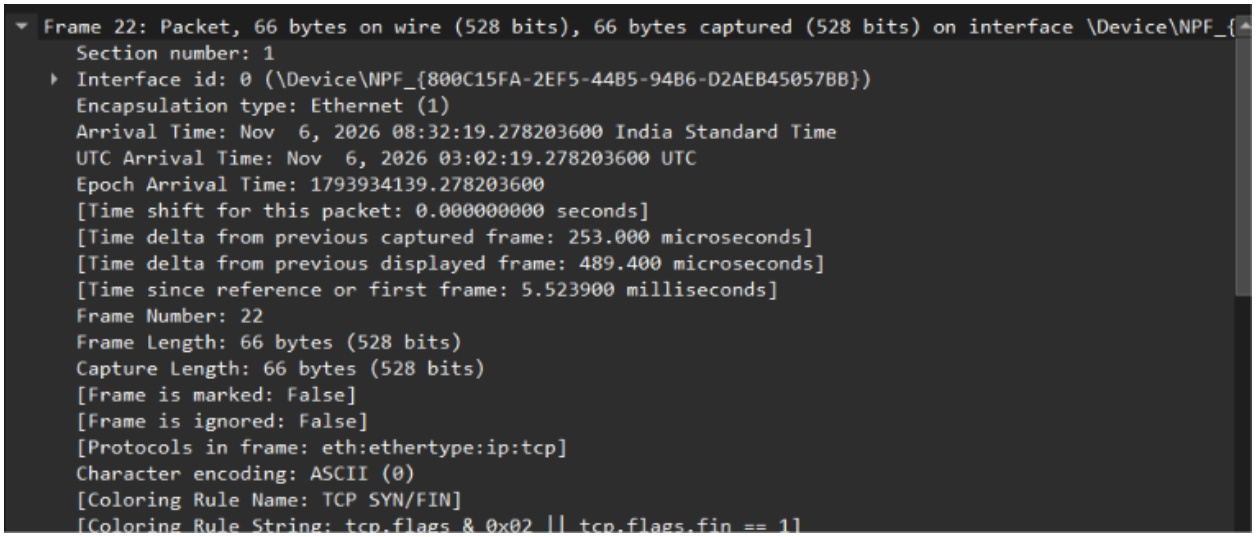
Tcp : 349 packets



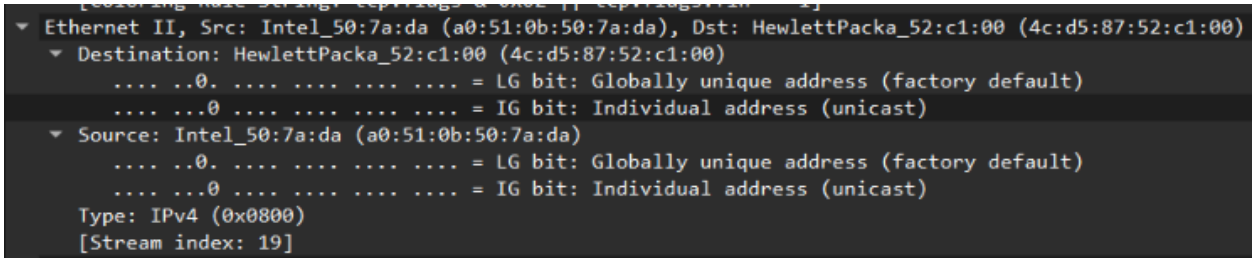
Udp: 14984 packets



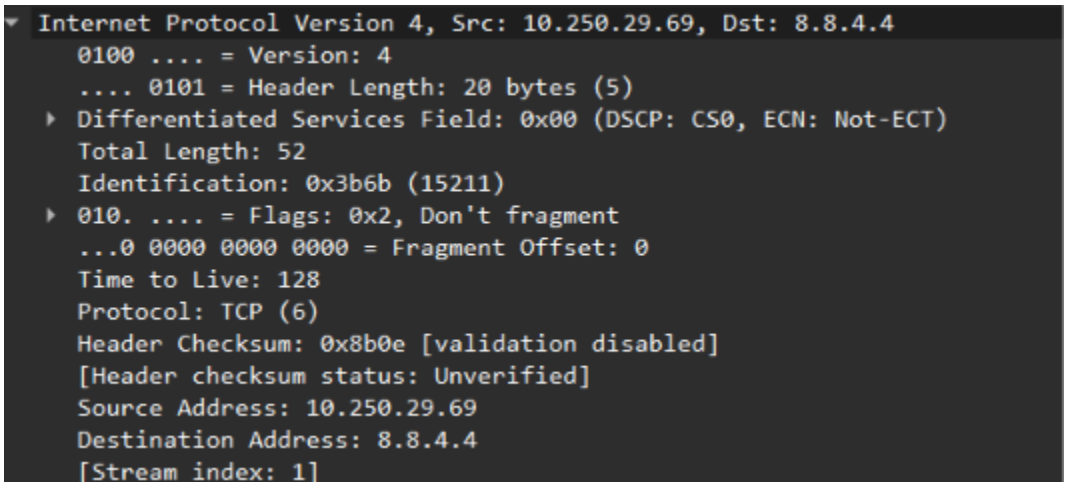
Icmp: 0 packets



Frame number: 22
Frame Length: 66



Protocol type : IPv4



Source ip: 10.250.29.69
Destination ip: 8.8.4.4
TTL: 128
Identification: 0x3b6b

Parameter	TCP Packet
Frame Length	66 bytes
Ethernet Header Length	14 bytes
IP Header Length	20 bytes

Transport Header Length (TCP)	32 bytes
IP Total Length	52 bytes
Transport Segment Length	32 bytes
Payload Length	0 bytes (if TCP header is 32 bytes)

Total size= E+ IP+ TCP+ Data

Total size = 14+20+32+0

Total size= 66 bytes

Total Overhead= E+ IP+ TCP

Total Overhead= 14+20+32

Total Overhead= 66 bytes

$$\text{Efficiency} = \frac{\text{Payload}}{\text{Total Size}} \times 100$$

$$= \frac{0}{66} \times 100$$

$$\boxed{0\%}$$

$$TCPPayload = TCPSegmentLength - TCPHeaderLength$$

1. What is encapsulation?

Encapsulation is the process of adding headers (and sometimes trailers) to data as it moves through different network layers before transmission.

2. What is the purpose of the Ethernet header?

The Ethernet header is used for communication within a local network. It contains:

- Source MAC address
- Destination MAC address
- EtherType (identifies the network-layer protocol).

3. What information is stored in the IP header?

The IP header contains:

- Source IP address
- Destination IP address
- IP version (IPv4/IPv6)
- Total length
- TTL (Time to Live)
- Protocol type (TCP/UDP).

4. What is the difference between TCP and UDP headers?

TCP Header	**UDP Header**
Minimum 20 bytes	Fixed 8 bytes
Contains sequence and acknowledgment numbers	Does not contain sequence or acknowledgment numbers
Provides reliable communication	Does not guarantee delivery

Has more control information	Has fewer fields

5. Why does each protocol add a header?

Each protocol adds a header to provide the information needed for communication, such as addressing, routing, error checking and reliable data delivery.

6. What is packet overhead?

Packet overhead is the extra data added to a packet in the form of headers and trailers. It is used for network communication and control rather than carrying actual application data.

7. How does header overhead affect network efficiency?

Header overhead reduces network efficiency because some transmitted bytes are used for control information instead of actual data. Larger headers mean less space is available for application data.

Example: If a packet contains 100 bytes of headers and 900 bytes of actual data, only 90% of the packet carries useful application data.